

Response to Reviewer #3 (Round 2) — Manuscript CSEM-D-25-02344R1

We thank the reviewer for pressing us on these three points. In the first round we substituted empirical/qualitative arguments for the formal analyses requested; this round, we have run the actual tests.

1. Formal tests for non-stationarity / structural breaks

We have added new Appendix A, "Formal Stationarity and Structural-Break Diagnostics," reporting:

- **Augmented Dickey-Fuller (ADF) and KPSS tests** on both the raw price level and the daily return series for all five assets. The pattern is completely consistent across every asset: the price level classifies as unit-root/non-stationary (ADF fails to reject its unit-root null; KPSS strongly rejects its stationarity null, $p \leq 0.01$), while the daily return series classifies as stationary (ADF rejects the unit-root null at $p < 0.001$ for every asset; KPSS fails to reject stationarity). This confirms our return-based modeling target is well-behaved even though the raw price process is not.
- **A CUSUM-of-OLS-residuals test** (Ploberger & Kramer, 1992) for mean-stability of the return series, and **a Quandt-Andrews Sup-F search** for the most likely single breakpoint date, with a parametric-bootstrap p-value (2,000 replicates, 15% trim). Neither test finds a statistically significant single mean-break in any of the five assets' return series at the 5% level (CUSUM $p \geq 0.21$; Sup-F bootstrap $p \geq 0.17$ in every case), even though the Sup-F search's candidate breakpoint dates land near well-known macro events (e.g. SPY: 2009-04-09, within days of the financial-crisis market bottom). We read this as supporting our framework's design premise: the non-stationarity we target is better characterized as continuous, gradual drift in predictability (consistent with Figure 1's reward-dispersion spikes) than as a small number of discrete regimes separated by sharp mean breaks — favoring an approach that continuously re-estimates the optimal predictor at every walk-forward window over one that would first need to detect and then react to isolated breakpoints.

We are explicit in the manuscript about what these tests can and cannot show: they formally establish non-stationarity at the level of price dynamics and test for (and do not find) a discrete break in the return series' mean, but they are not designed to, and should not be read as, direct evidence for the qualitative "regime shifts" invoked as motivation elsewhere in the paper. The absence of a discrete break is, if anything, more consistent with a picture of continuous, gradual drift in relative predictability and volatility than with abrupt regime changes — which is precisely the kind of non-stationarity our walk-forward CMAB framework is designed to track by

continuously re-estimating the optimal predictor, rather than a mechanism built to detect and react to discrete breakpoints.

We opted for ADF, KPSS, and the CUSUM/Sup-F family over Bai-Perron specifically because they are implementable directly from packages already in our dependency set (`statsmodels`), keeping the project's dependency surface unchanged; your comment framed ADF/KPSS/Sup-F/Bai-Perron as illustrative alternatives rather than a mandatory set, and we believe this combination directly answers the question asked.

2. Transaction-cost sensitivity

We have added new Appendix B, "Transaction-Cost Sensitivity," reporting annualized return, Sharpe ratio, and maximum drawdown for the CMAB-greedy and static-ensemble strategies under one-way transaction cost assumptions of 0, 5, 10, 25, and 50 basis points, applied whenever the strategy's binary long/cash position switches (not on every arm reassignment while the position itself is unchanged, since only a position switch corresponds to an actual trade in this single-instrument long/cash formulation). This is computed directly from the already-cached arm-selection sequences underlying Table 1, requiring no retraining. (An earlier draft of this letter described all three metrics as reported while the manuscript's table in fact only showed Sharpe ratio; the manuscript's Appendix B table now reports all three, as described below.)

Averaged across all 30 configurations, the CMAB-greedy strategy's zero-cost advantage erodes steadily as costs rise, for both Sharpe ratio (+0.14 at 0 bps, beating the static ensemble in 20/30 configurations) and annualized return (+0.05 at 0 bps, same 20/30 win count): 19/30 configurations still favor CMAB at 5 bps, 17/30 at 10 bps, and by 25–50 bps the average CMAB-minus-ensemble gap turns negative for both metrics, with only 14–15/30 configurations retaining an edge. Maximum drawdown shows a weaker, noisier pattern throughout (13–16/30 configurations favor CMAB depending on fee level, with an essentially nil average advantage even at zero cost), since drawdown depends more on the timing of a strategy's worst losses than on its average responsiveness.

To directly verify the mechanism, we separately computed each strategy's position-switch frequency (the fraction of out-of-sample days the long/cash position flips) from the same cached data. Averaged across all 30 configurations, CMAB-greedy switches position on 25.1% of days versus 19.4% for the ensemble (~30% more often), which is what produces the extra fee events. This is, however, policy-dependent, not a property of "the CMAB strategy" uniformly: LinUCB (24.7%) and especially Thompson Sampling (33.9%) switch substantially more than the ensemble, while Softmax is if anything slightly more conservative than the ensemble (16.8%). This is a genuine limitation for the more exploratory policies, reported plainly: at realistic institutional cost levels (single-digit to low-double-digit bps for the liquid instruments studied here) the CMAB framework retains its edge in the clear majority of configurations, but at retail-level costs (25–50

bps) the advantage is no longer guaranteed and should be assessed per-asset and per-policy, not assumed uniformly. Full per-configuration results for all three metrics and the position-switch frequencies are provided as `fee_sensitivity.csv` and `switch_rate.csv` in our GitHub repository, disaggregated by ticker, policy, and architecture, rather than reproduced in the manuscript in full.

We continue to view directly incorporating transaction costs into the CMAB's reward function (rather than as a post-hoc sensitivity analysis) as future work, as noted in our Conclusion — this fee-aware reward shaping would change the bandit's learned policy itself, not just the reported metrics, and is a substantively larger undertaking than the sensitivity analysis now included.

3. Statistical significance and multiple-testing correction

We appreciate the reviewer noting that our first-round response only addressed the seed-count half of this ask. We have now added new Section 4.4, reporting a paired Diebold-Mariano test and a block-bootstrap Sharpe-ratio-difference confidence interval, for every one of the 30 configurations underlying Table 1, using the same 25 independent seeds. Recognizing that 30 comparisons constitute a family for multiple-testing purposes, we apply both the Holm procedure (family-wise error rate, conservative) and Benjamini-Hochberg (false discovery rate) corrections at $\alpha = 0.05$.

0 of 30 configurations remain significant after Holm correction; 0 of 30 after Benjamini-Hochberg ($\alpha=0.05$). Only one configuration has a raw, uncorrected $p < 0.05$ (GC=F/Thompson/heterogeneous, $p=0.018$), and even that does not survive Holm correction (corrected $p=0.551$). A separate, non-multiplicity-corrected block-bootstrap Sharpe-difference confidence interval excludes zero for 2 of the 30 configurations, both favoring the CMAB strategy. We report this result plainly rather than selectively: the CMAB framework shows a consistent directional point-estimate edge (20 of 30 configurations favor CMAB, with the exceptions concentrated in TLT as already discussed in the manuscript), but individual per-asset gaps against the static ensemble do not clear a multiplicity-corrected significance bar given the sample sizes available (1,764 to 4,284 daily out-of-sample observations per configuration, depending on asset) and the serial correlation present in daily strategy returns. We have revised Section 4.1's "definitively dominant" language accordingly, and added an explicit statement to this effect in both Section 4.4 and the Conclusion. We report the primary test on the cross-seed-mean daily return against the deterministic static-ensemble baseline (the same aggregation Table 1's "greedy (mean)" column already uses) rather than treating each of the 25 seeds as an independent comparison, since the seeds share an identical market realization and an identical baseline and are therefore not independent replicates in the sense the correction procedure assumes. As a further robustness check, Section 4.4 also reports the same DM test run independently on each of the 25 seeds within every configuration (not fed into the multiple-testing correction, for the reason just given): the median configuration has 0% of its seeds individually rejecting at the uncorrected 5%

level, the mean is 5.9%, and no configuration has a majority of seeds rejecting — the per-seed evidence is consistent with, and does not contradict, the primary cross-seed-mean result.

We believe these additions directly close the three points raised and substantially strengthen the manuscript's empirical rigor.